

of the process parameters resulted in other products. It was sufficient for this purpose if it was shown that distinct differences existed in the properties of the products. This evidence might not include properties which could not be due to the product's substance parameters. See also T 279/84, T 151/95, T 728/98, T 564/02, T 1247/03.

In T 300/89 (OJ 1991, 480) it was stated that the application lacked novelty if the claim neither defined the product structurally nor mentioned all the specific conditions needed to obtain necessarily the products whose novelty could be demonstrated, e.g. by means of comparative tests. Similarly, in T 552/91 (OJ 1995, 100) the board ruled that the claim had to include all the process parameters required for unambiguously defining the claimed substances as inevitable process products. As chemical reactions only rarely took one particular course and therefore only rarely led to uniform substances, it was generally necessary to indicate not only the starting materials and reaction conditions, but also the method by which the reaction mixture was processed to obtain the claimed substances. This case law was confirmed in T 956/04, where the board considered that without clearly indicating the specific starting materials and the specific reaction conditions, the "obtainable-by" features as claimed failed to unequivocally define the claimed catalysts as inevitable process products.

In T 728/98 (OJ 2001, 319) the board stated that the general rule that the level of purity of a low molecular chemical compound could not entail novelty was also valid in the case of a product-by-process claim where the purity level was the inevitable result of the preparation process indicated in the claim. In the case in hand the appellant (applicant) had not been able to prove the existence of an exceptional situation where all attempts had failed to achieve a particular level of purity by conventional purification methods. See also chapter I.C.6.2.4. "Achieving a higher degree of purity".

In T 803/01 the board noted that there was no statement whatsoever in T 205/83 (OJ 1985, 363) which prohibited the presence of parameters relating to impurities in a claim for reasons of clarity. The parameters relating to the purity of the polylactides were technical features in line with R. 29(1) EPC 1973. This was considered to be a relevant criterion for assessing the extent to which the use of purity parameters in a product claim was allowable from the point of view of clarity (G 2/88, OJ 1990, 93).

In T 394/03 the board ruled that improved product quality achieved by means of a process did not generally constitute, in a product-by-process claim, a structural feature giving rise to novelty or inventive step. In the case in hand, a decorated ceramic product was claimed which differed from known products of this type only by virtue of its better quality, which had been achieved by means of the process according to the invention.

In T 564/02, when examining whether the product itself met the requirements for patentability, the board had to deal with the burden of proof. The respondent (opponent) had raised an objection of lack of novelty which was based on the disclosure in an example in a prior art document. The board observed that the validity of this objection rested exclusively on the validity of certain assumptions made by the respondent with respect to parameters of the product disclosed in prior art. In such a case, the concept of the balance of probabilities could not be applied for the assessment of the validity of each assumption;

it had to cede to a stricter criterion close to absolute conviction. In other words, there should be a degree of certainty which is beyond reasonable doubt.

7.3. Requirement that the claimed product cannot be described in any other way

The criterion laid down in T 150/82 (OJ 1984, 309), whereby it must be impossible to define the claimed product other than in terms of a process of manufacture is now established case law (see e.g. T 333/93, T 749/95, T 950/97, T 1074/97, T 933/01, T 150/12, T 863/12).

In T 320/87 (OJ 1990, 71) the board stated that product-by-process claims were admissible for hybrid seeds and plants when they were not individually definable biological entities which could be characterised by their physiological or morphological features.

In T 130/90 the board had to rule on recombinant monoclonal antibodies having two different specificities and produced by a process involving trioma cells or quadroma cells and thus producing antibodies in a native form. The patentability of the process was not questioned. The process known from prior art chemically recombined antibody half-molecules. At issue was the validity of the product claim. The prior art did not disclose how fully re-associated molecules, i.e. with the same structure as native antibodies, which might be contained in the mixture of hybrids, could be screened out and isolated from molecules which were chemically altered. The board allowed a claim for bispecific recombinant monoclonal antibodies comprising intact immunological chains produced by the process claimed in the independent process claims of the patent in suit. Defining the antibodies by their process was the only way of delimiting them vis-à-vis the prior art.

In T 552/91 (OJ 1995, 100) it was held that where a European patent application relates to chemical substances originally defined by an incorrect chemical structural formula, correction of which is not allowable under R. 88 EPC 1973, replacement of the incorrect formula by the correct one infringes Art. 123(2) EPC 1973. Nevertheless, the submission of a "product-by-process" claim is compatible with Art. 123(2) EPC 1973 if it contains all the measures required to obtain this result (starting materials, reaction conditions, separation).

7.4. Applicability of the principles for product-by-process claims to use claims

In T 81/14 the board held that when considering the definition of a product in terms of its production process the principles developed in the jurisprudence for the product-by-process claims are in general to be applied, also in the case of a claim directed to the use of that product. Claim 1-7 of one of the auxiliary requests were directed to a method of producing a sintered cemented carbide body. Claim 8 was directed to the use, for the production of a cutting tool, of a sintered cemented carbide body obtained according to the method of any one of claims 1-7. Hence, claim 8 comprised process and product features and was notionally equivalent to a claim directed to a process for the production of a cutting tool using the sintered cemented carbide body. Although the claim was not directed to a product but to a process, the board stated that the principles underlying the assessment of the clarity of product features should not depend on whether such product

features appear in a claim directed to a product or in a claim directed to a process. Applying the principles developed in the jurisprudence for product-by-process claims to claim 8, the board stated it would have been possible to define the sintered body in terms of structural features. Accordingly, the use of a product-by-process definition in claim 8 resulted in a lack of clarity.

7.5. Combination of product and process features

In decision T.148/87 the board stated that it was admissible to combine product parameters and process parameters in the same claim. In T.129/88 (OJ 1993, 598) the board was of the opinion that the inclusion in a product claim of one or more process features might be permissible if their presence was desirable having regard to the impact of the national laws of one or more contracting states (see also T.592/95, T.288/02). See also in this chapter II.A.3.7.

7.6. Extension of protection conferred by product-by-process claims

In T.411/89 the board had to decide whether the amendment of a product-by-process claim from "obtained" to "obtainable" extended the protection conferred by the patent. The board took the view that the protection was not extended because the amendment did not modify the definition of the product which was claimed per se from the beginning and because the process used for its characterisation remained the same.

In T.423/89, by restricting the claim to only one of a number of manufacturing processes specified in the original claim and disclosed in the description, the patent proprietors had ceased to claim absolute product protection and had undertaken a significant limitation of their claim. There were therefore no objections under Art. 123(3) EPC 1973. The change in category from a product-by-process claim to a manufacturing process claim was also admissible in this case, as the protection afforded by the granted patent had to extend to all those methods of manufacture covered by the processes described in the claim and disclosed in the patent specification.

In T.20/94, however, objections under Art. 123(3) EPC 1973 led to the board's refusing a change from process claims to a product-by-process claim by way of amendment, as despite the fact that a product-by-process claim was characterised by the process for its preparation, it nevertheless belonged to the category of claim directed to a physical entity and was a claim directed to the product per se. The scope of protection conferred by a product claim exceeded the scope conferred by a process claim under Art. 64(2) EPC 1973. See also chapter II.E.2.7. "Change of claim category".

8. Claims fees

Until 31 March 2008, under R.31(1) EPC 1973 and then the new R.45 EPC, any European patent application comprising more than ten claims incurred payment of a claims fee in respect of each claim over and above that number. With effect from 1 April 2008 the number of fee-exempt claims was increased to fifteen. For international applications

entering the European phase the corresponding provisions are R. 110 EPC 1973 and R. 164 EPC.

The Legal Board ruled in case J 9/84 (OJ 1985, 233) that R. 31(1) EPC 1973 was to be interpreted as meaning that claims numbered 1 to 10 upon filing of the European patent application – or transmittal to the EPO of the international application under the PCT – were exempt from fees, and that claims numbered 11 onwards were not. Accordingly, abandoning a fee-exempt claim after filing – or transmittal – did not have the effect that its fee exemption was transferred to another claim.

8.1. Number of claims

The question has arisen on a number of occasions as to whether part of, or an annex to, the description of a European patent application comprising a series of numbered paragraphs drawn up in the form of claims should be regarded as claims for the purposes of R. 31(1) EPC 1973.

In J 5/87 (OJ 1987, 295) the application contained, under the heading "Claims", ten claims; however it contained 33 further claims which were annexed to the description as preferred, numbered embodiments presented in the form of claims. The addendum appeared in form as well as in substance only to consist of claims within the meaning of Art. 84 and R. 29 EPC 1973. Furthermore, the appellant had admitted that the addendum had been included in the application in order to maintain the possibility of making its content a basis for substantive examination. As a result, the applicant was required to pay the claims fee – applicants were not at liberty to ignore the provisions on the form and content of the description and claims contained in R. 27 and R. 29 EPC 1973.

In J 15/88 (OJ 1990, 445) the Legal Board decided in a similar case that although the 117 disputed "clauses" in question were numbered and arranged as claims and seemed to define matter in terms of technical features, they were not to be regarded as such since the fact remained that they were never referred to as claims and there were claims elsewhere which were referred to as such.

Decisions J 16/88, J 29/88, J 25/89, J 26/89, J 27/89, J 28/89, J 34/89 and T 490/90 all confirmed this opinion, pointing out that the case dealt with in J 5/87 differed from the others since the applicant had indicated an intention to regard the annexes as claims. The intention of the appellant not to have this part considered as claims is more important than the form of the text.

8.2. No refund of claims fees

In J 3/09 the Legal Board rejected the appellant's request for a refund of the claims fee for claims 16-68 and noted that the legal basis for payment of those claims was R. 110(1) EPC 1973. It added that fees, once validly paid, were not refundable unless provision to the contrary was made (see also J 11/12). The appellant had not relied on any specific provision of the EPC in its support. In response to the appellant's argument that it was inevitable that the subject matter of the unsearched claims could not be prosecuted

in the application, and had therefore to be considered as abandoned, the board noted *inter alia* that in the frequent case of an *a posteriori* objection of non-unity, even if justified, the claims could usefully serve as a basis for later limiting the claimed subject-matter to a more specifically defined but now unitary and searched invention. Whilst the extent to which this was possible depended on the circumstances of the individual case, it remained the position that the payment of claims fees for claims exceeding the number of ten could have a useful purpose on its own independently of the question whether all the claims concerned unitary subject-matter.